



Temperature and precipitation variability in eastern Anatolia: Results from independent component analysis of Lake Van sediment data spanning the last 250 kyr BP

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ABSTRACT

In this study, we present the results of independent component analysis (ICA) of previously published Lake Van data covering the last 250 kyr BP, to shed light on the precipitation and temperature regime in eastern Anatolia. The data processed were the element intensities of Ca, Fe, K, Mn, and Si analyzed by XRF core scanner; concentrations of TOC and CaCO₃ content; and B* (color reflectance) from the Ahlat Ridge sediment record. Our analysis is based on application of ICA on the data by changing the initial random unit vector several times and clustering possible independent components through average-link agglomeration. As components extracted by ICA do not have a hierarchy, mutual information, which is a measure of information content between two random variables, is used as a measure of similarity by which to select candidate components. As a result, we argue, the independent component (Van-IC8), which shows the highest similarity to the Greenland $\delta^{18}\text{O}$ record and visually similar to other regional temperature indicating data, can be read as a proxy for temperature variability. We also assert that the independent component (Van-IC7) which has the highest similarity to B* and visually similar to other regional precipitation proxies, and with the lake level reconstruction from another sediment profile from Lake Van is a proxy of precipitation variability across the region. Our results show that the region's temperature approximately maps onto global records, i.e. warm during interglacials and cold during stadials. However, the precipitation proxy reveals that the region was not dry, or at least as wet as it is today, during the end of the MIS 6 and the LGM. The MIS 5e/c and Holocene were characterized by a wet period followed by dry intervals and Dansgaard/Oeschger events are characterized as being warm and wet.

1. Introduction

Proxy measurements are indirect indicators of past environments and many studies use multi-proxy approaches to overcome the shortcomings of single proxy data. One reason for studying proxy data is to produce a simplified model of past environments. In this study, we use Independent Component Analysis (ICA) as a parameter extraction tool by which to analyze published sediment data from Lake Van (Kwiecien et al., 2014; Stockhecke et al., 2014b, 2016). The results of our analyses can be argued to reflect temperature and precipitation variability across eastern Anatolia. While previous models based on almost the same data were not able to capture lake level changes, where it was claimed as these changes were shaped by precipitation/evaporation balance (Kuzucuoğlu et al., 2010). We claim that the models presented in this study have the ability to explain these geomorphological features seen around the lake.

In 2010, the PALEOVAN (hereafter PV) drilling campaign in Lake Van was conducted as an ICDP project. In order to study past climatic and environmental changes, a 220 m long composite profile was taken (see Fig. 1 for the location). Subsequent laboratory analyses constitute the longest and most complete paleolimnic archive in Anatolia. This dataset has been used in several studies with the aim to understand various aspects of the paleolimnic state (see the special issue: Litt and Anselmetti, 2014). The PV results show the presence of several glacial/interglacial cycles within the Lake Van sediment data (see for example Litt et al., 2014; Kwiecien et al., 2014; Randlett et al., 2014) and reveal that Lake Van is a unique natural laboratory for paleoclimate studies.

Previous studies, which are not related to PV, only provided continuous proxy data for the last 20 kyr BP at most (Landmann et al., 1996a; Kempe et al., 2002; Wick et al., 2003). Field studies conducted by Valetton (1978), Landmann et al. (1996a), Kempe et al. (2002) and Kuzucuoğlu et al. (2010) revealed the presence of several terraces at

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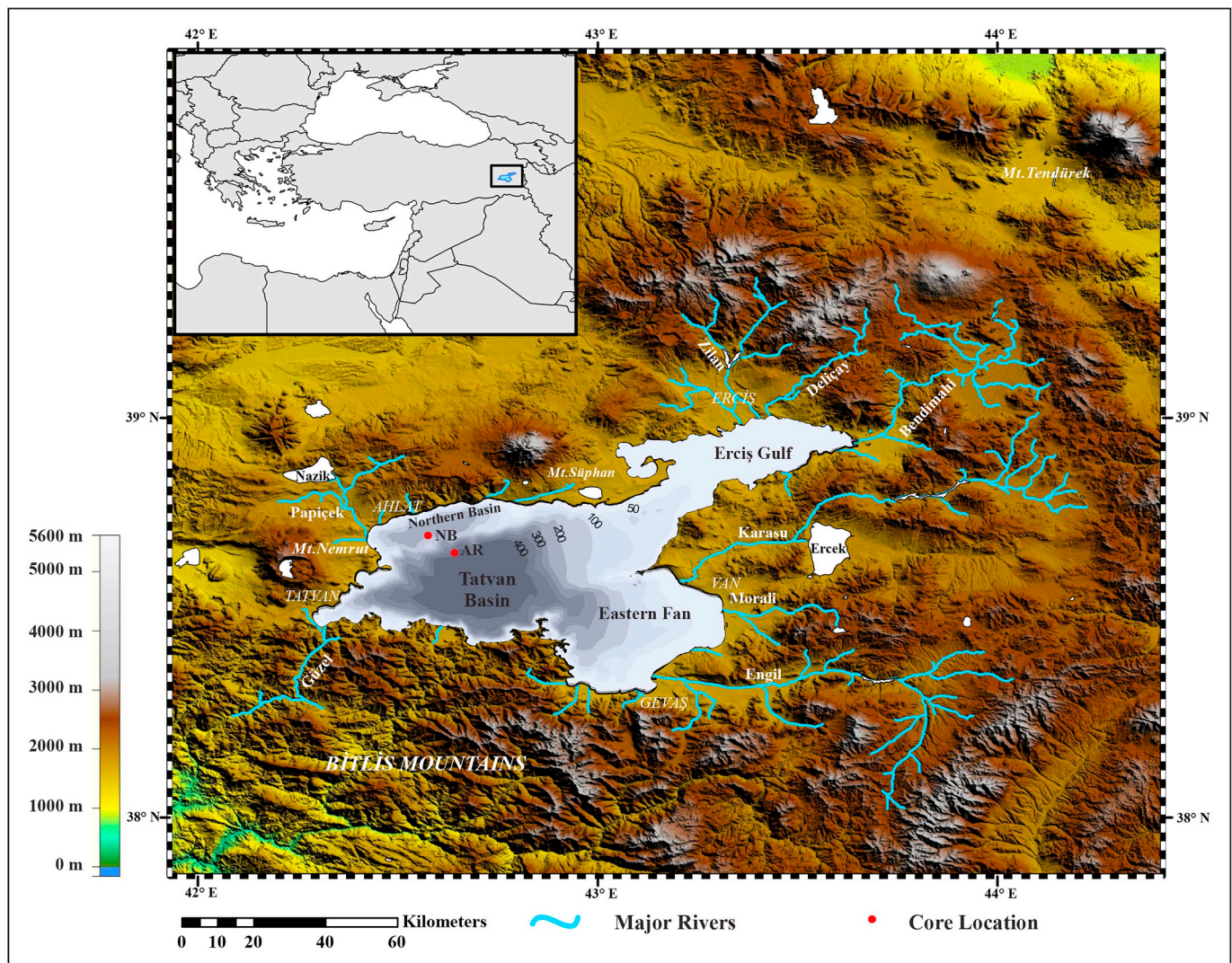


Fig. 1. Location map of Lake Van (modified from Ön et al. (2017b)). Red dots indicate the location of the Ahlat Ridge (AR) and Northern Basin (NB) composite profiles retrieved by the PV project. Major mountains and the locations of meteorological stations mentioned in the text are indicated. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

different topographic heights around Lake Van with terraces that stand between 20 m and 80 m higher than the present-day lake level and which have been dated to between 26 and 20 kyr BP - the Last Glacial Maximum (LGM) - and which have been interpreted as evidences of a wet LGM in the region (Landmann et al., 1996a; Kuzucuoğlu et al., 2010). However, results from previous studies and new data from the PV suggest different interpretations. Some recent analyses using PV data show that the period was drier than the present and that lake level was 200 m below its current level during the LGM (Stockhecke et al., 2016; Tomonaga et al., 2017). The claim by Stockhecke et al. (2016), that the region was dry during the LGM, is mainly based on principal component analysis (PCA) of AR data, but through this study, we discuss the reasons why the results of PCA were inappropriately evaluated by Stockhecke et al. (2016) and offer a new proxy-based model for climate variables of eastern Anatolia using ICA.

ICA is a state of the art signal extraction tool with its roots in information theory (Hyvärinen and Oja, 2000). ICA's main algebraic difference from data reduction techniques based on second order statistics, such as principal component analysis (PCA), is that, rather than maximal decorrelation, ICA rotates the axes in directions that maximize statistical independence. More specifically, PCA aims to find orthogonal

components, ordered according to the variance that they explain and removes linear correlations between data. ICA is, by definition, a signal separation method and has the ability to remove higher order dependence. In the field of paleoclimate multiproxy studies, it is reasonable to pursue a principal component analysis approach if one is almost confident that data variability is governed by a known agent, or to eliminate noise in the data and then conduct aimed analyses. It is also possible to combine the two former ideas to remove the shared variance based on solid assumptions via genuinely designed tests (e.g. see Bloemsma et al. (2012)). On the other hand, the leading component obtained through applying PCA on a matrix containing a highly correlated vector set will be a resembling copy of those correlated vectors, where the analyst should be aware. Furthermore, Gaussianity of the stochastic variables is a strict constraint of PCA and since it depends on variance-covariance matrix and outliers should be removed through statistical means before analysis. On the other hand, an ICA approach might be suitable for the problem under consideration because individual independent components, in theory at least, correspond directly to observable quantities, such as precipitation or temperature. It is designed to extract independent processes from linearly mixed signals. For this reason, ICA has recently been applied in a paleoclimate

study of Lake Hazar (Ön et al., 2017a) in which paleoenvironmental chemistry proxies were assumed to be mixtures of different independent processes and were accordingly used as a basis by which to extract paleo-precipitation and paleo-temperature variability. ICA has also been applied as a signal separation tool on other subjects such as chemistry-based “multiproxy” datasets, similar to paleoclimate data, from different disciplines such as medicine (Krumstiek et al., 2012) or applied geochemistry (Yang and Cheng, 2015; Iwamori et al., 2017). It has also been applied in geographical sciences such as hydrology (Westra et al., 2007; Middleton et al., 2015) and climate science (Aires et al., 2000; Hannachi et al., 2009). ICA works on two major assumptions that are constraints on the results. These are non-Gaussianity and statistical independence of aimed components. For the purpose of this study, it is assumed that precipitation and temperature are non-Gaussian random variables (for an extensive discussion, see Perron and Sura, 2012) that are assumed to be quasi-independent, at least for the studied long term interval, with the characteristics of climate variables changing dynamically through non-linear interactions.

In this study, ICA is applied on element intensities (Ca, Fe, Mn, K, Si) from XRF-core-scanning analysis (Kwiecien et al., 2014), total organic carbon (TOC) and carbonate content (CaCO_3) (Stockhecke et al., 2014b) and sediment reflectance color data in blue-yellow chromaticity (B^*) (Stockhecke et al., 2016). Two of the extracted components are interpreted as proxies of temperature and precipitation variability in the region based on their statistical similarity to Greenland $\delta^{18}\text{O}$ data, operating through the assumed dynamic atmospheric relation between the North Atlantic and the eastern Mediterranean (Wegwerth et al., 2015). Results are also supported by other regional and global paleo-data. The temperature record roughly follows the variability observed in Greenland NGRIP $\delta^{18}\text{O}$ data, such as the fact that glacials are relatively cold, interglacials are relatively warm, and abrupt warming events of MIS 3 are observable. The precipitation record also reflects abrupt warming events observed in Greenland records, i.e. the region is shown to be relatively wet during Dansgaard/Oeschger events. However, the end of the MIS 6 and LGM appear to be wet, unlike in the PCA results of Stockhecke et al. (2016), which used Si, Ca, Ti, TOC and B^* data. Our results for precipitation closely follow lake level reconstruction of Çağatay et al. (2014) and support the high-stand lake level during the LGM, as the observed terraces also appear to confirm.

2. Regional setting

Lake Van fills a tectonically active depression at eastern Anatolia and is surrounded by a number of semi-active volcanoes to the north and east (Yılmaz et al., 1998). The Bitlis Mountains, which are a part of the southeastern Taurides, separate the basin from the lowlands of the Arabian plate and mark the transition between Mediterranean and continental climate zones (Türkeş, 1996; Ünal et al., 2003). This manifests itself in the region's flora, with the region being on the boundary between forest and steppe-type biomes (van Zeist and Bottema, 1982). According to available meteorological data, average precipitation around the lake is variable with annual average precipitation amounts (data collected and organized by Göktürk, 2005) as follows: i) 380 mm/yr at the east of the lake (Van meteorological station), ii) 580 mm/yr at the southeast (Gevaş station), iii) 800 mm/yr at the southwest (Tatvan station), iv) 560 mm/yr at the west (Ahlat station), v) 440 mm/yr at the north (Erciş station).

The current eastern Mediterranean winter precipitation regime is mainly under the influence of the Atlantic Ocean, with cyclones that originate over the Mediterranean Sea and Asiatic dry high pressure systems. Which one of those factors is the most important in shaping the precipitation regime of the eastern Mediterranean, and of Anatolia in particular, is still a subject of debate. However, the general consensus favors (Türkeş and Erlat, 2003) the so-called North Atlantic Oscillation, which is a pressure gradient regime, for the winter months. On the

other hand, dry and cold air systems originating in Asia may be fed by temperate Black Sea surface waters to cause precipitation over Black Sea coasts in winter (Bozkurt and Şen, 2011; Göktürk et al., 2011). During summer, the factors dominating the region's climate are the subtropical high pressure system and the low pressure trough settling over Persian Gulf due to the Indian monsoon. Even though the mean summer surface pressure is lower than the mean winter surface pressure over the Aegean and Mediterranean coasts and the winds that flow over the sea are moisture laden, summers are almost dry in western Anatolia (Bozkurt et al., 2012). However, the Black Sea coasts of Anatolia are, again, an exception, receiving precipitation by the monsoonal low circulation saturated with warm Black Sea waters. Therefore, the precipitation regime of the Anatolian peninsula as a whole is far from straightforward to describe, lacking a single pressure system as it does.

In the present day, the lake stands approximately 1650 m above sea level and has a surface area of 3570 km². Water salinity is ~ 21.4‰ and water pH is ~ 9.8 (Reimer et al., 2009). During the winter, Mediterranean humid air loses moisture as precipitation over the colder orographic southern boundaries of Anatolia and therefore cannot feed the region. Furthermore, the geographical proximity of eastern Anatolia to cold/dry anti-cyclones from the Asian interior results in a relatively drier regional climate during the winter. Therefore, unlike the dominant winter precipitation in Mediterranean coastal zones, precipitation in eastern Anatolia is mainly recorded in spring and at the end of fall (Figure S1). Winter precipitation is mainly snow and summer in the region is almost completely dry. Seasonality indices of monthly precipitation data show that the region is on the border of Continental Mediterranean and Continental Eastern Anatolian climates types (Türkeş, 1996). The seasonality of climate parameters controls the sedimentation process and varve formation in the lake. Increased runoff during spring and fall provides dissolved calcium ions, detritus, and organic material which manifest as light layers in sediment profiles. Stratification of the lake waters in summer increase organic productivity in the lake, and it also contributes to the formation of light layers. In winter surface waters cool and runoff decreases; consequently stratification ceases and suspended organic matter precipitates, forming the dark layers of sediment profile as a result (Lemcke and Sturm, 1997; Stockhecke et al., 2012).

3. Materials and methods

3.1. Lake Van Ahlat Ridge sediment record and chronology

The 220 m long Ahlat Ridge (AR) composite profile was drilled at a depth of 360 m below the present lake level with the primary aim of understanding the climatic evolution of the region (Litt and Anselmetti, 2014).

The AR record has been subjected to XRF elemental scanning using an Avaatech core scanner by Kwiecien et al. (2014) for every 2 cm over a 1 cm² area with a slit size of 12 mm. The results are collated by 10 s sampling time and with 10 kV and 0.2 mA generator settings. Also, the color profile of the sediment in L*A*B* color space was photographed at 0.03 mm resolution (Stockhecke et al., 2016). CaCO_3 and TOC measurements were established for every 2.5 cm through the record (Stockhecke et al., 2014b).

While the age model of the uppermost part of the AR record was constructed by ¹⁴C datings and correlation to previously gathered robust varve chronology (Landmann et al., 1996b), the chronology of the remainder of the record was established by tuning the proxy data to: i) NGRIP $\delta^{18}\text{O}$ GICC05 and GICC05modelext timescale (Andersen et al., 2004; Steffensen et al., 2008; Svensson et al., 2008; Wolff et al., 2010) for the interval between 0 and 116 kyr BP and, ii) speleothem based and EDC3 based synthetic Greenland record (Barker et al., 2011) for the interval between 116 and 400 kyr BP and 400 and 650 kyr BP, respectively. This age model was then cross-referenced with several other

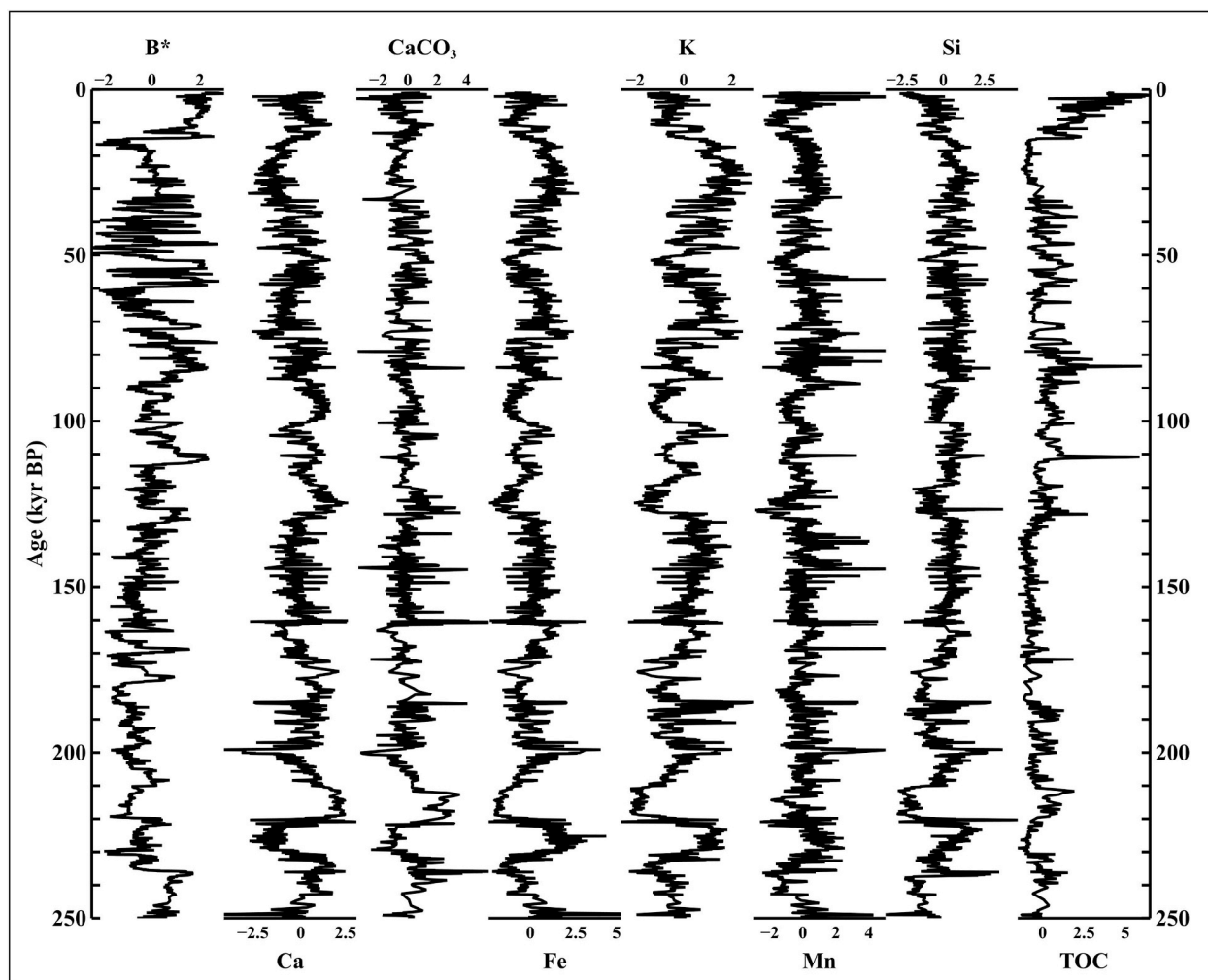


Fig. 2. Data processed in this study after interpolated to the age model of Fe and Mn, as described in Section 3.1. All the data in this study have been standardized by making each data zero mean and unit variance.

dating methods, including $^{40}\text{Ar}/^{39}\text{Ar}$ datings, tephrostratigraphy, magnetostratigraphy and ^{10}Be measurements. The model was then improved by using the AICC2012 timescale (Veres et al., 2013) for the interval between 90 and 125 kyr BP and by adding more tie points (Stockhecke et al., 2016). Whereas the difference between the old and improved age models is negligible for the period younger than 100 kyr BP, before 100 kyr BP there exist intervals with significant age differences, reaching almost 10 kyr (for all the details and construction process of the chronology, see Stockhecke et al. (2014a), Stockhecke et al. (2014b) and Stockhecke et al. (2016)). The complete record represents the last 600 kyr. In this study, we use only the last 250 kyr section of the data due to the discontinuities below this point.

The resolution of the published XRF data by Kwiecien et al. (2014) and Stockhecke et al. (2016) and B^* , TOC, CaCO_3 data are different and therefore we linearly interpolated the data to the one with fewer data points, which corresponds to the data presented in Kwiecien et al. (2014). Before interpolation, we applied a first order Savitzky-Golay filter (Savitzky and Golay, 1964) with a 5 window length to CaCO_3 , TOC, Ca, K, Si and with a 53 window length to B^* . After the smoothing process, every 1 in 5 elements in the former dataset and every 1 in 53 elements in B^* were selected to bring the data to approximately the same number of rows as Fe and Mn. Afterwards, filtered and resampled data were linearly interpolated to the age model's (developed by Stockhecke et al., 2016) of Fe and Mn from Kwiecien et al. (2014) (Fig. 2).

3.2. FastICA

FastICA is a numerical implementation of ICA that has been widely used during the last decade for problems as diverse as telecommunications theory, face recognition, image processing (see Hyvärinen et al., 2004, Part 4), hydrology (Westra et al., 2007), climatology (Hannachi et al., 2009), and, recently, in paleoclimatology (Ön et al., 2017a).

FastICA is one of the major blind signal separation methods with some critical assumptions. It has distinct advantages when compared to methods based on variance-covariance matrix, such as PCA or factor analysis. Its most important advantage is that it is not restricted to the second-order statistical constraints brought in by the quadratic nature of covariance. Secondly, extracted independent components have no hierarchy among themselves, unlike principal components which are based on the eigenvalue spectrum. However, not being subject to any hierarchy may be seen as a disadvantage because interpretation of the resulting components, at least for this study, depends on similarity methods, as set out below.

The FastICA model assumes that the matrix of observed variables $\mathbf{X}$ (in our case, the Lake Van AR record) is a linear transformation of the source matrix $\mathbf{S}$ (matrix of independent components) by the mixing matrix $\mathbf{A}$, which is;

$$\mathbf{X} = \mathbf{AS}. \quad (1)$$

Here, neither the mixing matrix $\mathbf{A}$, nor the source matrix $\mathbf{S}$ is

known. Since, the sole object is to find the source matrix, the method tries to estimate $\mathbf{W}$, which is the inverse of $\mathbf{A}$, and finds $\mathbf{S}$ by;

$$\mathbf{S} = \mathbf{W}\mathbf{X}. \quad (2)$$

In order to establish that this is mathematically possible, ICA approach assumes the following:

1. The source components that are being looked for should be statistically independent,
2. Source components should have non-Gaussian distributions, and
3. Mixing matrix $\mathbf{A}$ should be invertible.

The non-Gaussianity assumption here, through the central limit theorem, is the key to estimating the independent components (Hyvärinen and Oja, 2000).

FastICA searches for independent components through the well-defined, information theoretical concept of negentropy. The definition of negentropy of a random vector $\mathbf{y}$ is,

$$J(\mathbf{y}) = H(\mathbf{y}_{\text{gauss}}) - H(\mathbf{y}), \quad (3)$$

where H is the Shannon's entropy and defined over the probability mass function P of $\mathbf{y}$ as:

$$H(\mathbf{y}) = - \sum_{\mathbf{y} \in \mathbf{Y}} P(\mathbf{y}) \log P(\mathbf{y}). \quad (4)$$

Intuitively speaking, negentropy defines a distance between the real distribution of a random variable and the normally distributed random variable with the same mean and variance.

Negentropy of a random vector defined by eq. (3) is theoretically calculated by probability mass functions of random variables as given by eq. (3). However, to describe the probability mass function of a random variable is a non-trivial problem. In order to estimate negentropy, FastICA uses a method based on maximum entropy principle with a selected contrast function (Hyvärinen, 1998; Hyvärinen and Oja, 2000). The contrast function is the nonlinear component of the analysis and for this study, we used:

$$G(u) = \log(\cosh(u)), \quad (5)$$

which Hyvärinen (1999) stated is suitable for general purposes such as this.

The FastICA scheme for estimating the first independent component is based on a selection of a random unit vector and execution of a fixed point iterative process until convergence is established, as described in Hyvärinen and Oja (2000). The same algorithm used in estimating the first component is then used to estimate the rest of the components, by the deflation method. However, to prevent the convergence of components into the same directions outputs are decorrelated after every iteration through each estimation, as explained in Hyvärinen and Oja (2000).

Documented examples in Brys et al. (2005) show that the worst outliers in the data may affect the results of FastICA. Here the term “worst outliers” (used by Brys et al. (2005)) is taken to mean outliers that are not defined through classical spherical boundaries, whereas they are defined by the term *medcouple* for skewed data and preserve the non-Gaussian nature and skewness of the data. We applied the adjusted outlyingness method (Hubert and Van der Veeken, 2008) to eliminate the outliers for multivariate data by using the LIBRA library (Verboven and Hubert, 2010) in MATLAB, which is a generalization of the method described in Brys et al. (2005). The results from our case show that marking approximately 1.8% of all observations as outliers and deleting them before the FastICA operation is sufficient. Having deleted 89 data-rows, 4905 remain.

Before starting the FastICA algorithm, a random unit vector is selected. However, this selected random vector may slightly affect the results of FastICA, so Himberg et al. (2004) suggest to changing the initial vector several times using the Icaasso package, which was

originally developed for neuroimaging applications. After running the algorithm several times, the method projects the results onto a 2-D plane by curvilinear component analysis (Demartines and Herault, 1997) and clusters the components gathered in each run via group average-link agglomeration, to check for the reliability of each component, prior to computing the distances of each candidate component from other candidates in the cluster and offering the component with the highest summed similarity value in each cluster. In order to validate the results of FastICA, we ran the algorithm 250 times by changing the randomly selected initial point with the contrast function described at eq. (5), through the Icaasso package. Unlike PCA, there is no need to standardize the data to apply FastICA (however, they need to be centered and whitened by transforming to orthogonal unit vectors, which is an essential preliminary step to remove the second-order dependency on the data). In each way the results will be the same, however, we chose to standardize the data before the analysis.

One peculiarity of ICA in our context is that, unlike other well-known dimension reduction methods (such as PCA or FA), neither the variances nor the order of the components are known (Hyvärinen and Oja, 2000). Therefore, a strategy should be chosen by which to select appropriate components. For example, in a similar study, Ön et al. (2017a) decided the contextually appropriate components by looking at distance correlation coefficients (Székely et al., 2007) between the extracted components and well-known climate proxies. This distance correlation approach may fail to satisfy equitability (Reshef et al., 2011; Kinney and Atwal, 2014) meaning that, among different type of relationships with same amount of noise, the statistic should give a similar score. Alternatively, we can use another quantity called mutual information for this purpose, since the data have enough elements to almost define probability mass functions by using histograms. Mutual information is a measure which finds the amount of information shared between two systems by using the Shannon's entropy idea in information theory (Cover and Thomas, 2006) and it is theoretically equitable (Reshef et al., 2011). Mutual information between $\mathbf{x}$ and $\mathbf{y}$ is defined as:

$$I(\mathbf{x}, \mathbf{y}) = \sum_{\mathbf{x} \in \mathbf{X}} \sum_{\mathbf{y} \in \mathbf{Y}} p(\mathbf{x}, \mathbf{y}) \log \frac{p(\mathbf{x}, \mathbf{y})}{p(\mathbf{x})p(\mathbf{y})} \quad (6)$$

where $p(\mathbf{x})$, $p(\mathbf{y})$ are the marginal probability mass functions and $p(\mathbf{x}, \mathbf{y})$ is the joint probability mass function of the random variables $\mathbf{x}$ and $\mathbf{y}$. Notice that $I(\mathbf{x}, \mathbf{y}) = 0$ if and only if $\mathbf{x}$ and $\mathbf{y}$ are statistically independent. Without loss of generality, we used e as the base of the logarithm through computations of mutual information.

Polar ice core $\delta^{18}\text{O}$ data provide reliable and almost continuous sources for past climate change and they are good reflectors of paleotemperature change in particular (Joussaume and Jouzel, 1993). In order to select the appropriate components, as a preliminary approach we computed the mutual information between each resulting independent component and each proxy record used through FastICA and also NGRIP $\delta^{18}\text{O}$ data (Andersen et al., 2004). Note that the NGRIP $\delta^{18}\text{O}$ data is padded with synthetic Greenland data (Barker et al., 2011), and have been linearly interpolated to the AR age model. While computing the mutual information, we described the marginal and joint probability functions by constructing histograms with 50 fixed width bins between the minimum and maximum value of the data. Mutual information does not give comparable results like the results of Pearson's correlation coefficient between $[-1, 1]$. However, Joe (1989) proposes a transformation method of mutual information results as

$$I^*(\mathbf{x}, \mathbf{y}) = \sqrt{1 - \exp(-2I(\mathbf{x}, \mathbf{y}))}. \quad (7)$$

By doing so, mutual information results are transformed between 0 and 1 (Table 1).

4. Results and discussion

Using the methodology described by Himberg et al. (2004) (see

Table 1

Transformed mutual information results between the independent components of the AR record and NGRIP $\delta^{18}\text{O}$ combined with synthetic Greenland record and the AR data. Mutual information is measured in nats.

I^*	Van-IC1	Van-IC2	Van-IC3	Van-IC4	Van-IC5	Van-IC6	Van-IC7	Van-IC8
NGRIP- $\delta^{18}\text{O}$	0.33	0.39	0.36	0.41	0.37	0.48	0.51	0.54
B*	0.45	0.44	0.44	0.42	0.35	0.56	0.72	0.49
Ca	0.26	0.45	0.29	0.34	0.38	0.42	0.50	0.74
CaCO_3	0.32	0.76	0.37	0.36	0.30	0.29	0.42	0.48
Fe	0.30	0.45	0.32	0.35	0.48	0.42	0.51	0.73
K	0.26	0.44	0.29	0.34	0.34	0.43	0.57	0.80
Mn	0.33	0.37	0.37	0.62	0.35	0.35	0.44	0.46
Si	0.34	0.48	0.30	0.40	0.35	0.38	0.48	0.78
TOC	0.60	0.54	0.42	0.36	0.03	0.50	0.36	0.39

Figure S2, S3, S4 and S5 for plots of different methods used to select the appropriate number of independent components), there are eight statistically valid clusters, i.e. it is proper to say that there are eight independent processes governing the eight dimensional data-set and correspondingly eight independent components (see Figure S6). Since the aim is to select climate-related components, we followed a preliminary strategy of selecting candidates by calculating a statistical similarity measure (mutual information in our case) of the independent

components with the Greenland (NGRIP) stable oxygen isotope record (Andersen et al., 2004) combined with the synthetic Greenland isotope record (Barker et al., 2011).

Transformed mutual information results indicate that (Table 1), Van-IC8 give the highest similarity with the Greenland data. Furthermore, Van-IC8 shows visual similarity with the Greenland record (Fig. 3) and with PC1 constructed by Stockhecke et al. (2016) (principal component which has the highest variance, see Fig. 5 therein), where

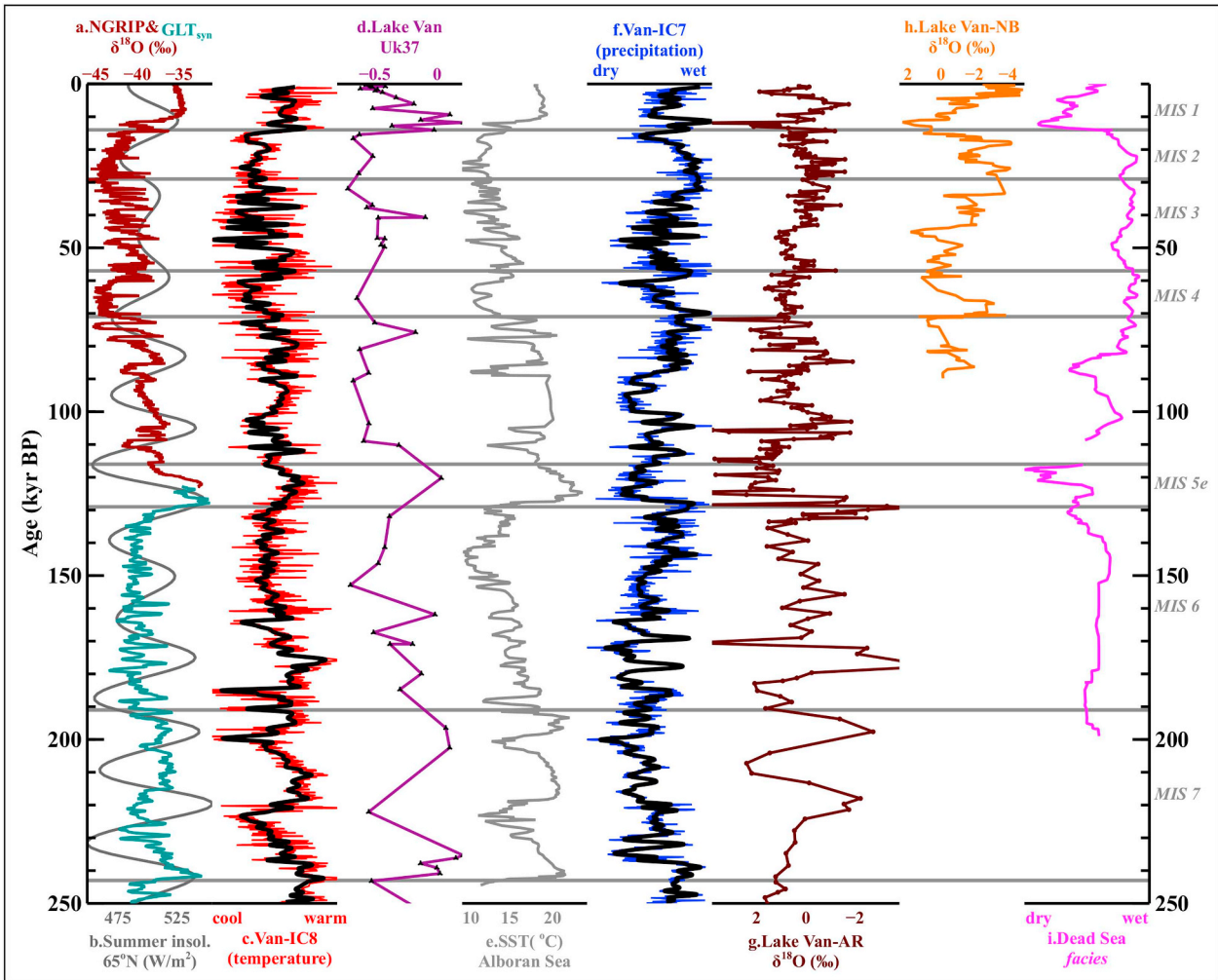


Fig. 3. Comparison of selected independent components with global and regional records. a. NGRIP $\delta^{18}\text{O}$ data (Andersen et al., 2004) combined with synthetic Greenland record (Barker et al., 2011). b. Summer insolation for 65°N (Berger and Loutre, 1991). c. Temperature proxy of Lake Van Van-IC8, proposed in this study. d. Lake Van Uk37 index, as a proxy for temperature (Randlett et al., 2014). e. Alboran Sea, surface temperature (Martrat et al., 2004). f. Precipitation proxy of Lake Van Van-IC7, proposed in this study. g. Lake Van AR $\delta^{18}\text{O}$ data (Kwiecien et al., 2014). h. Lake Van NB $\delta^{18}\text{O}$ data (Çağatay et al., 2014). i. Reconstruction of the hydroclimate of Levant by aragonite-detritus laminae (Torfstein et al., 2015). Black curves over Van-IC7 and Van-IC8 are Butterworth lowpass filter designed with 1.5 kyr cutoff period applied to IC's interpolated to 50 years evenly spaced data. The horizontal lines are the MIS boundaries defined in Lisiecki and Raymo (2005).

PC1 was interpreted as hydroclimatic variability through the sole assumption of B^* , Si, K, TOC and Ca being sensitive to hydroclimate variations. The interpretation of Stockhecke et al. (2016) associating the hydroclimatic variability to the first principal component may seem reasonable, since maximal variance in sediment records of a lake should be mainly driven by precipitation, evaporation and surface/groundwater runoff components. However, in the AR record, linear correlation between Ca, K and Si are exceptionally high (Figure S7) and this can be seen in Fig S1 in Stockhecke et al. (2016). Therefore, PC1 of Stockhecke et al. (2016) is mainly governed by the variances in K, Ca and Si (represented by the loadings of PCA, see Fig S3 in Stockhecke et al. (2016)) constituting a significant bias in their analysis. The bias stems from the fact that their principal components are constructed using only five different time series and three of them are highly correlated. Similar results can be obtained if PCA is applied to the extended data used in this study (see Figure S9), since four of the eight profiles are highly correlated (see Figure S7). We therefore reject the interpretation of PC1 in Stockhecke et al. (2014a) and elaborate on this below. On the other hand, Randlett et al. (2014) constructed a curve based on alkenone Uk37 index (Fig. 3) and claim that it reflects changes in temperature and haptophyte species composition. Visual similarity between Van-IC8 and low resolution Uk37 index is high, and the reader should note that the age model used by Randlett et al. (2014) is based on an older chronology that is different from the most recent one up to 10 kyr during some intervals of MIS 6 (cf. Stockhecke et al. (2014a) and Stockhecke et al. (2016)). Van-IC8 also resembles changes in sea surface temperature of the western Mediterranean (Martrat et al., 2004) and Black Sea (Wegwerth et al., 2015). The similarity of Van-IC8 to Greenland isotope data, Uk37 index of Lake Van, and sea surface temperature data (Figs. 3 and 4) allows us to state that it reflects the paleo-temperature variability for the region. Using the Greenland $\delta^{18}\text{O}$ record as a paleothermometer is a subject of debate (see Brook (2013); Vinther and Johnsen (2013) and references therein). However, it remains the only commonly used high resolution hemispheric paleo-temperature proxy. At this point, the relation between Van-IC8 and the data used in ICA can be queried but the high linear correlation between Ca, Fe, K and Si (Figure S7 and S8) make it difficult to compare and discuss Van-IC8 with the proxy data. Therefore, our interpretation of the Van-IC8 as a temperature proxy stands only on its overall similarity to the NGRIP $\delta^{18}\text{O}$, Uk37 index and sea surface temperatures from the wider region. According to this temperature variability, the region's temperature approximately followed the hemispheric temperature curve, i.e. cold during glacial/stadials and warm during interglacials/interstadials.

The second highest similarity with the Greenland isotope data is established by Van-IC7 (Table 1). The Van-IC7 curve very closely resembles the Greenland curve (Fig. 3), especially for abrupt events. However, trends and/or levels are not the same as in the case for the claimed temperature proxy, Van-IC8. The most striking differences are seen during times of maximal glaciation (the end of Late Pleistocene and MIS 6) and during interglacials, especially through MIS 1, MIS 5e and MIS 5c.

Lake Van sediments are laminated and these laminations are described as varves by Kempe and Degens (1978). Annual lamina couplets consist of dark and light alternations (Landmann et al., 1996b; Stockhecke et al., 2012). According to Stockhecke et al. (2012), light layers represent the effects of spring runoff, summer organic productivity and mixing and runoff in fall. While autochthonous biological productivity partly depends on the temperature, its quantity is largely determined by runoff through the transfer of organic matter, dissolved nutrients and mineral particles (Zolitschka et al., 2015). Unstratified water column in Lake Van results in dark colored layers during winter. However, it must be noted that changes in precipitation variability may also affect the thickness of dark layers (Stockhecke et al., 2012). Therefore, it can be seen that annual meteorological conditions, but mainly precipitation, are reflected in laminated sediments which in turn

mainly define the changes in color profile of the sediment data. In a region where seasonality of precipitation is high (see Section and Figure S1), the color of sediments mainly reflects precipitation variability (Putnam and Broecker, 2017). According to mutual information results, Van-IC7 has the highest similarity with normalized B^* data (Table 1), which shows the highest information transferred to Van-IC7 through the analyzed data. In light of the above discussions, we claim that Van-IC7 is a precipitation variability proxy for the region. Although the precipitation model presented here does not agree with the hydroclimate variability and lake level reconstruction presented in Stockhecke et al. (2016), Van-IC7 does show good correlation with lake level curve of the last 90 kyr BP (Fig. 5) constructed by Çağatay et al. (2014).

Further supporting evidence for our interpretation of Van-IC7 comes from the $\delta^{18}\text{O}_{\text{bulk}}$ data (Fig. 3) retrieved from two different Lake Van composite profiles - AR (Kwiecien et al., 2014) and NB (Çağatay et al., 2014). The isotopic composition of carbonates in Mediterranean lakes is dependent on a number of different factors (Roberts et al., 2008). However, Lemcke and Sturm (1997) and Roberts et al. (2008) explicitly state that the primary controlling factor of isotope fractionation in Lake Van is evaporation/precipitation balance. Kwiecien et al. (2014) (similarly Çağatay et al. (2014)), after an extensive discussion reach a *tentative interpretation* (the original authors' term) about the $\delta^{18}\text{O}$ variability of Lake Van carbonates. They claim that $\delta^{18}\text{O}$ is a proxy of temperature and isotopic condition of the epilimnion subject to the precipitation type/season in the region. This conclusion was reached by rejecting the proposition that relative humidity was the main controlling factor of $\delta^{18}\text{O}$ and by assuming that glacial periods should be drier than interglacials. By assuming that relative humidity is the main controlling agent for $\delta^{18}\text{O}$ fractionation, the isotope profiles follow the trend of Van-IC7 during each marine isotope stage (readers should also note that the age model of $\delta^{18}\text{O}$ profiles depend on the chronology constructed by Stockhecke et al. (2014a)). Correlation between the isotope profiles and Van-IC7 is not perfect, which is probably a result of dependence of $\delta^{18}\text{O}$ on other factors as stated by Roberts et al. (2008). There are also differences between the NB and AR isotope profiles but general trends in each marine isotope stage do resemble one another and the difference between them may be a result of site difference, different age models and/or different laboratory measurements.

As already stated in Section 1, earlier continuous records from Lake Van are not long enough to test the results presented in this study. However, these studies suggest that the LGM should be considered to have been wet, judging by the terraces around the lake, where the highest ones are 80 m above today's level (Kuzucuoğlu et al., 2010). On the other hand, neither the hydroclimate model (namely, PC1) nor the lake level reconstruction based on a low resolution climate model (Timmermann et al., 2014) presented by Stockhecke et al. (2016) capture a wet period during the LGM. According to the lake level reconstruction (where it has been expressed as an idealized low resolution experiment by the authors), Stockhecke et al. (2016) suggest it should have been somewhere around 200 m below today's level during the LGM. This means that existing proposed models fail to demonstrate an almost 280 m lake level rise. However, while most high resolution paleoproxies of Lake Van published by Kwiecien et al. (2014) and Stockhecke et al. (2016) correctly recover abrupt changes in the North Atlantic, such as Dansgaard-Oeschger events, PC1 fails to show a geologically established lake level rise of nearly 280 m. Furthermore, there is also a higher resolution hindcast climate simulation with a more realistic precipitation pattern (Menviel et al., 2014) for the period between 50 kyr and 11 kyr BP which is presented in Stockhecke et al. (2016) but not used in lake level modeling, also it does not show low rates of precipitation during the LGM (Fig. 4). Hence, the arguments of Stockhecke et al. (2016) about the lake level, clearly contradict the results of terrace studies.

Other than Lake Van, there are lakes with high-stand water levels throughout the LGM not only in Anatolia (Fontugne et al., 1999;

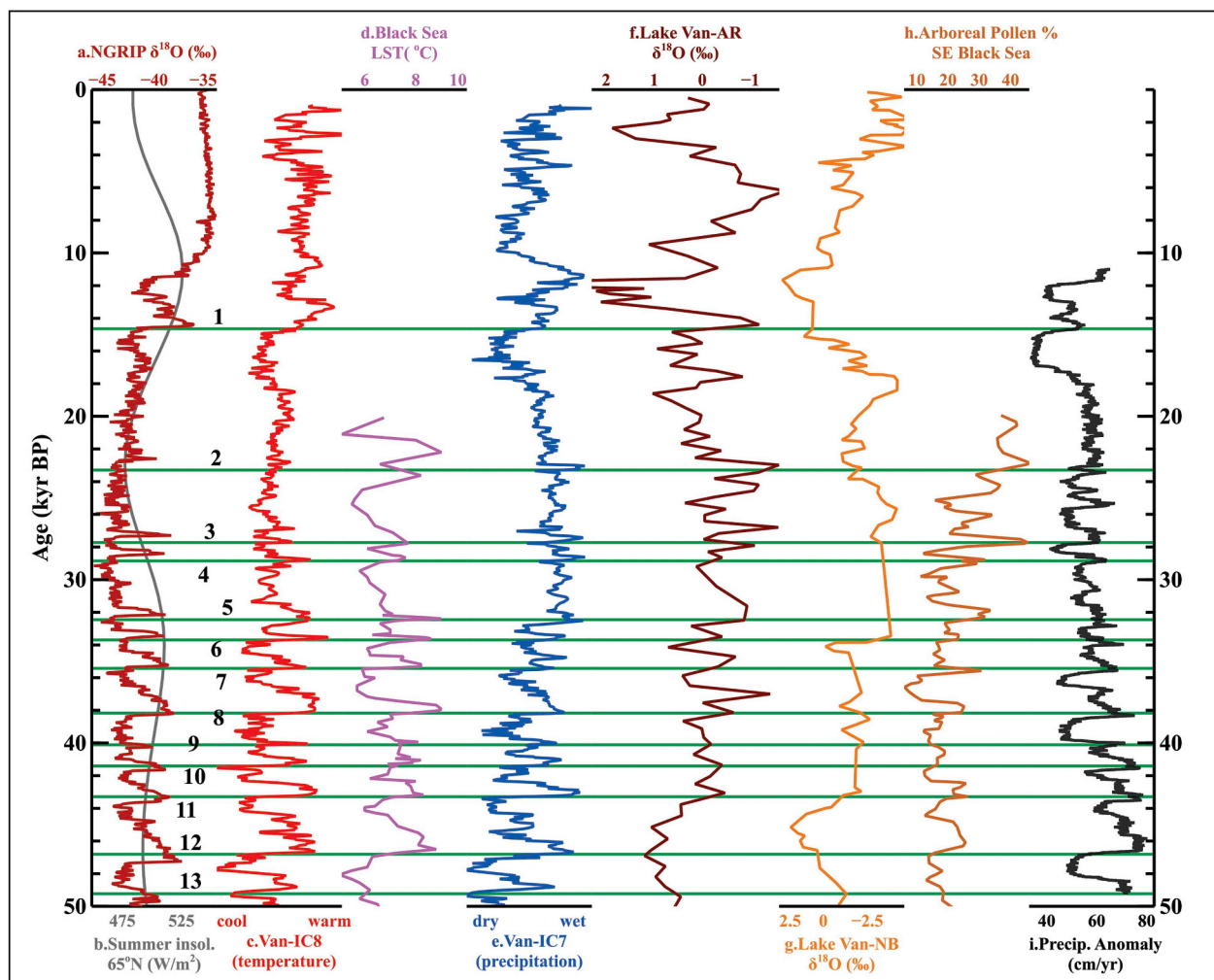


Fig. 4. Comparison of selected independent components with global and regional records for the last 50 kyr. a. NGRIP $\delta^{18}\text{O}$ data (Andersen et al., 2004). b. Summer insolation for 65°N (Berger and Loutre, 1991). c. Temperature proxy of Lake Van Van-IC8, proposed in this study. d. Black Sea, lake surface temperature (Wegwerth et al., 2015). e. Precipitation proxy of Lake Van Van-IC7, proposed in this study. f. Lake Van AR $\delta^{18}\text{O}$ data (Kwiecien et al., 2014). g. Lake Van NB $\delta^{18}\text{O}$ data (Çağatay et al., 2014). h. Arboreal pollen percentage from core 25-GC1 (Shumilovskikh et al., 2014), south Black Sea. i. LOVECLIM model precipitation anomaly data (Menviel et al., 2014) for Lake Van region presented in Stockhecke et al. (2016). Green lines indicate the start of the Greenland interstadials (Svensson et al., 2008). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

Kuzucuoğlu et al., 1999) but also in the Middle East (Torfstein et al., 2013), southern Europe, and north Africa (Prentice et al., 1992; Harrison and Digerfeldt, 1993; Harrison et al., 1996). There are two schools of thought about the reasons for these high lake levels in the eastern Mediterranean during the LGM (Tzedakis, 2007). The first maintains that, during the LGM, very cold winters and cool cloudy summers caused a decrease in evaporation (COHMAP Members, 1988). However, according to the water-balance model of past lake level changes and paleovegetation cover in Lake Ioannina, this hypothesis does not hold up, yet it should be a wet period (Prentice et al., 1992). Moreover, sequences of thick aragonite-detritus deposits around the Dead Sea appear to imply that evaporation rates were in fact higher than they are today, at least in the Dead Sea basin (Torfstein et al., 2013). Despite higher evaporation, the Dead Sea lake level was significantly higher than the present day (Torfstein et al., 2013). Furthermore, Torfstein et al. (2015) claim that the end of MIS 6 should have been wet and the end of MIS 5e should have been dry according to the Dead Sea facies record (see Torfstein et al. (2015) due to the particular way in which they define the term facies). Within the age uncertainties in both records, Van-IC7 appears to follow the pattern of the Dead Sea facies curve (Fig. 3), supporting a climate-induced high-stand for Lake Van during the LGM. On the other hand, while pollen data from Lake Van do not reflect any increase in precipitation during LGM (Pickarski et al.,

2015), there are other pollen records from across the Anatolian peninsula that so show evidence for high precipitation rates. Low percentage rate of arboreal pollen in the Lake Van profile during the LGM may be a result of severe colds and/or day-night diurnal difference across the region which is typical of high altitude geographies. Pollen profiles from the Marmara Sea (Mudie et al., 2002) and Black Sea (Shumilovskikh et al., 2014, Fig. 4), regions that are not affected by the continentality that Lake Van experiences, reveal low temperatures associated with moderate precipitation. Furthermore, glacier advances in south/southwest Anatolia during the LGM have been explained by high precipitation rates by one-dimensional ice flow models in two studies (Sarkaya et al., 2008, 2014), both of which claim that for the observed glacial advances to have formed in south Anatolia during the LGM, precipitation should be almost twice its current rate.

High precipitation around the end of MIS 6 and the LGM is likely to be the result of the expansion of the Northern Hemisphere ice sheet (Harrison and Digerfeldt, 1993; Torfstein et al., 2013). The strengthening of the polar anticyclone may have forced the extratropical low pressure and subtropical high pressure systems to migrate southward and cause the Van region to be wet at this time, although a climatic link between the North Atlantic and eastern Anatolia is more direct during Greenland interstadials, which have been shown to be relatively warm and wet at the region (Fig. 4).

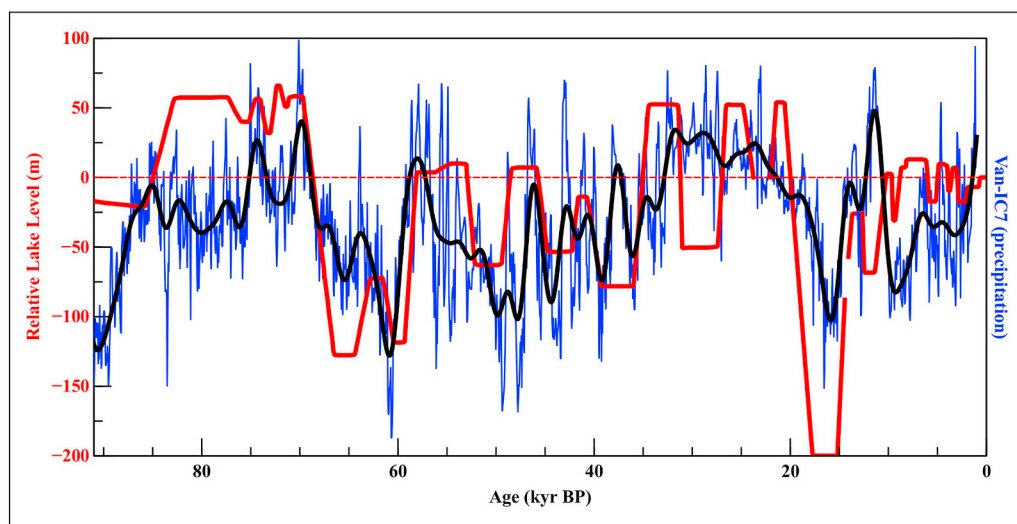


Fig. 5. Lake level changes constructed by Çağatay et al. (2014) are plotted in red and Van-IC7 in blue and Butterworth lowpass filter of Van-IC7 with cutoff period 2.5 kyr in black. Note that, lake level reconstruction of Çağatay et al. (2014) depends on lake terraces, seismic stratigraphy and lithology of a different composite record (NB) drilled from Lake Van. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

5. Conclusions

In this study, we offer the results of independent component analysis of the Lake Van AR data as possible reflection of temperature and precipitation variability of eastern Anatolia during the last 250 kyr.

Our methodology was to first remove outliers in the data using an adjusted outlyingness method, then apply FastICA by changing the initial random vector several times and cluster the results. Preliminary selection of the climate related variables depends on mutual information measure between the results of FastICA and NGRIP $\delta^{18}\text{O}$ data. Consequently, we suggest that two of the independent components can be interpreted as to reflect temperature (Van-IC8) and precipitation (Van-IC7) variability of the region. Our suggestions about these selected components are supported by proxy data from Lake Van and other regional records that also claimed to reflect temperature/precipitation variability.

According to our results, the Lake Van temperature record maps onto the NGRIP $\delta^{18}\text{O}$ record, whereby interglacials and interstadials are relatively warm and stadials are relatively cold. However, precipitation follows a different pattern, visually resembling the Dead Sea records and Lake Van level changes. During times when glaciation is at its maximum (the end of MIS 6 and LGM), precipitation increased. During interstadials (MIS 5e/c and Holocene) a wet early stage was followed by a dry period.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.quaint.2018.11.037>.

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